POLITE NUMBERS AND ODD DIVISORS

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Abstract. A positive integer is called polite if it is a sum of two or more consecutive positive integers. We give a short self-contained proof that the impolite numbers are exactly the powers of two, count the representations of an integer as a sum of consecutive positive integers in terms of its odd divisors, and record a few consequences.

1. Introduction

Writing an integer as a sum of a block of consecutive positive integers, as in $9 = 4 + 5 = 2 + 3 + 4$, is a classical recreational topic; the count of such representations goes back at least to Mason [2]. The purpose of this note is to give a complete and short account organised around a single factorisation identity (Lemma 2.2). Our main result is Theorem 2.6, characterising the polite numbers. In Section 3 we count representations, and in Section 4 we combine the main theorem with Bertrand's postulate.

2. Representations and the main theorem

Definition 2.1. Let $n \geq 1$ be an integer. A *representation* of n is a pair (a, k) of positive integers such that

$$n = a + (a+1) + \cdots + (a+k-1) = \sum_{i=0}^{k-1} (a+i).$$

The integer k is the *length* of the representation. Every n has the *trivial* representation $(n, 1)$. We call n *polite* if it has a representation of length $k \geq 2$. We write $R(n)$ for the number of representations of n, $R^*(n)$ for the number of representations of n of length at least 2, and $d_{\mathrm{odd}}(n)$ for the number of odd positive divisors of n.

Lemma 2.2. *Let n, a, k be positive integers. Then (a, k) is a representation of n if and only if $2n = k(2a + k - 1)$.*

Proof. By the formula for an arithmetic progression, $\sum_{i=0}^{k-1}(a+i) = ka + \frac{k(k-1)}{2} = \frac{k(2a+k-1)}{2}$. □

Lemma 2.3. *Let a, k be positive integers. Then k and $2a + k - 1$ have opposite parity, and $k < 2a + k - 1$.*

Proof. The sum $k + (2a + k - 1) = 2a + 2k + 1$ is odd, so exactly one of the two numbers is odd. Their difference is $(2a + k - 1) - k = 2a - 1 \geq 1$. □

Lemma 2.4. *Let $n \geq 1$. If n has a representation of length $k \geq 2$, then n has an odd divisor greater than 1.*

Proof. Let (a, k) be such a representation and put $m = 2a + k - 1$. By Lemma 2.2, $2n = km$, and by Lemma 2.3 exactly one of k, m is odd; call it d. Since the other factor is even, d divides n. If $d = k$ then $d \geq 2$, hence $d \geq 3$ as d is odd. If $d = m$ then $d > k \geq 2$. In both cases $d > 1$. □

Proposition 2.5. *Let $n \geq 1$ and suppose that n has an odd divisor $d \geq 1$. Then n is polite.*

Proof. Write $n = dm$ with $m \geq 1$. If $2m > d$, put $k = d$ and $a = m - \frac{d-1}{2}$; then $a \geq 1$, and $k \geq 3$ because d is odd and $d \geq 3$. If $2m < d$, put $k = 2m$ and $a = \frac{d-2m+1}{2}$, a positive integer since d is odd; here $k \geq 2$. (The case $2m = d$ is impossible as d is odd.) In either case $k(2a + k - 1) = 2dm = 2n$, so (a, k) is a representation of n of length at least 2 by Lemma 2.2. □

Theorem 2.6 (Main theorem)**.** *Let $n \geq 1$ be an integer. Then n is polite if and only if n is not a power of 2, that is, if and only if $n \neq 2^j$ for every integer $j \geq 0$.*

Proof. If $n = 2^j$, then the only odd divisor of n is 1, so n is not polite by Lemma 2.4. Conversely, if n is not a power of 2, write $n = 2^j m$ with m odd; then $m \neq 1$, so m is an odd divisor of n greater than 1, and n is polite by Proposition 2.5. □

3. Counting representations

Proposition 3.1. *For every $n \geq 1$ we have $R(n) = d_{\mathrm{odd}}(n)$.*

Proof. Send a representation (a, k) to the odd member d of $\{k, 2a + k - 1\}$, an odd divisor of n as in the proof of Lemma 2.4. Conversely, given an odd divisor d of n, put $e = 2n/d$, which is even, and let $k = \min(d, e)$ and $a = \frac{\max(d,e) - \min(d,e) + 1}{2}$. Then $a \geq 1$ and $k(2a + k - 1) = de = 2n$. By Lemma 2.3 the two maps are mutually inverse. □

Corollary 3.2. *For every $n \geq 1$ we have $R^*(n) = d_{\mathrm{odd}}(n)$.*

Proof. Every representation other than the trivial one $(n, 1)$ has length at least 2, and $(n, 1)$ is the only representation of length 1. Hence the claim follows from Proposition 3.1. □

Corollary 3.3. *For every integer $j \geq 0$ we have $R(2^j) = 1$.*

Proof. The only odd divisor of 2^j is 1; apply Proposition 3.1. □

Lemma 3.4. *For every integer $k \geq 1$, the triangular number $T_k = \frac{k(k+1)}{2}$ is polite.*

Proof. We have $T_k = 1 + 2 + \cdots + k$, so $(1, k)$ is a representation of T_k of length k. □

Proposition 3.5. *Let p be a prime with $p \geq 5$. Then $R(p) = 2$, the two representations being $(p, 1)$ and $\left(\frac{p-1}{2}, 2\right)$.*

Proof. Since p is odd, its odd divisors are 1 and p, so $R(p) = 2$ by Proposition 3.1. Both listed pairs are representations, because $\frac{p-1}{2} + \frac{p+1}{2} = p$ and $\frac{p-1}{2} \geq 1$. □

4. An application of Bertrand's postulate

We use Bertrand's postulate, proved by Chebyshev [1], in the following form: *for every integer $n \geq 1$ there is a prime p with $n < p < 2n$.*

Corollary 4.1. *For every integer $n \geq 2$ the open interval $(n, 2n)$ contains a polite number.*

Proof. Let p be a prime with $n < p < 2n$. Then $p > 2$, so p is odd and not a power of 2; hence p is polite by Theorem 2.6. □

Remark. By Corollary 3.3 and Proposition 3.1, $R(n) = 2$ holds exactly when the odd part of n is prime.

Conjecture 4.2. *There are infinitely many n with $R(n) = R(n + 1) = 2$.*

References

[1] P. L. Chebyshev, *Mémoire sur les nombres premiers*, J. Math. Pures Appl. **17** (1852), 366–390.

[2] T. E. Mason, *On the representation of an integer as the sum of consecutive integers*, Amer. Math. Monthly **19** (1912), 46–50.

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